

Arin Bhandari

475-251-9074 | arinb@andrew.cmu.edu | [linkedin.com/in/arin-bhandari](https://www.linkedin.com/in/arin-bhandari) | arin.bhandari.us

EDUCATION

Carnegie Mellon University

Aug. 2023 – Dec. 2026

B.S. in Computer Science, Concentration in Computer Systems

Pittsburgh, PA

- Grade Point Average: 4.0/4.0, Dean's List with High Honors
- Courses: Compiler Design, Distributed Systems, Computer Networks, Machine Learning, Algorithm Design and Analysis
- Activities: Science Olympiad President, Student Senate Finance Chair, Carnegie Mellon Racing

EXPERIENCE

Databricks

May 2026 - Present

Software Engineer Intern

Mountain View, CA

- Building a performance modeling tool for Dicer, Databricks' auto-sharding service, that replays real production workloads to benchmark and iterate on sharding algorithms offline, eliminating production risk

Carnegie Mellon School of Computer Science

Jan. 2024 - Present

Teaching Assistant

Pittsburgh, PA

- Led grading sessions and collaborated with over 40 other teaching assistants to host weekly lab sessions and office hours for two classes, introductory data structures and distributed systems, each taken by 500+ students a semester
- Developed a mark-and-sweep garbage collector assignment and Autolab grading infrastructure for C0, a memory-safe subset of C, supporting reliable memory management for student-written programs and automating testing and feedback

Amazon

May 2025 - Aug. 2025

Software Development Engineer Intern

New York, NY

- Built LLM-driven feature extraction systems to identify rare, predictive signals from unstructured product titles across 10B+ products, leveraging concept recognition and TF-IDF for high-signal attribute discovery
- Engineered automated AWS pipelines to process and extract signals from 100B+ SKUs, improving scalability and reliability of large-scale data workflows

Palantir

Mar. 2025 - Mar. 2025

Palantir Launch Intern

Washington D.C.

- Chosen as 1 of 40 students nationwide for Palantir's highly-selective Launch program, focusing on Palantir's real-world applications of data engineering and AI

AutoNation

Jun. 2024 - Aug. 2024

AI Developer Intern

White Plains, NY

- Engineered advanced search functionality using AWS Bedrock and OpenSearch, enabling vector embeddings and natural language queries to navigate a 300K+ car inventory, reducing response time by 25% for better user experience

PROJECTS

C0 Compiler

- Built a compiler for C0, a memory-safe subset of C, in OCaml targeting LLVM and x86 assembly
- Implemented advanced optimizations such as Global Value Numbering (GVN), Function Inlining, Aggressive Dead Code Elimination (ADCE), and Sparse Conditional Constant Propagation (SCCP), achieving speedup on par with GCC -O1

Distributed Bitcoin Miner

- Implemented a reliable, in-order messaging protocol over UDP using sliding windows, acknowledgments, checksums, and exponential backoff to tolerate packet loss and reordering
- Built a distributed Bitcoin mining system on top of the protocol, coordinating parallel workers and detecting node failures via heartbeats and epoch-based timeouts

Raft Consensus Algorithm

- Implemented the Raft consensus protocol in Go, including leader election, heartbeats, log replication, and commit indexing for a replicated state machine
- Designed concurrency-safe RPC handlers and background goroutines to ensure correctness under message loss, reordering, and node failures

Mixnet (Privacy-Preserving Overlay Network)

- Built a privacy-preserving overlay network combining encryption, randomized routing, and batching-based mixing to resist traffic analysis
- Evaluated system behavior across multiple network topologies using STP convergence time, routing message volume, and end-to-end RTTs on distributed testbeds

TECHNICAL SKILLS

Languages: Java, Python, Go, C, C++, Scala, OCaml, SQL, TypeScript, Assembly x86

Libraries: Pandas, NumPy, OpenCV, PyTorch, TensorFlow

Developer Tools: REST API, AWS, Git, Docker